

Solving multi-step equations



1 Consider the equation $9(n + 5) = 4n + 50$ for $n = 1$.

- a Find the value of the left-hand side of the equation when $n = 1$.
- b Find the value of the right-hand side of the equation when $n = 1$.
- c Consider your answers from parts (a) and (b).
Is $n = 1$ a solution of $9(n + 5) = 4n + 50$?

2 Consider the equation $0.28n - 0.4 = 2.3n - 3.03$ for $n = 0.8$.

- a Find the value of the left-hand side of the equation when $n = 0.8$.
- b Find the value of the right-hand side of the equation when $n = 0.8$.
- c Consider your answers from parts (a) and (b).
Is $n = 0.8$ a solution of the equation $0.28n - 0.4 = 2.3n - 3.03$?

3 Determine whether the given value of the variable is a solution to the equation.

a $p = 0; 3p - 9 = 7p - 1$

b $x = -2; 3x - (4x - 8) = -6$

c $x = \frac{7}{12}; x - \frac{4}{9} = \frac{17}{36}$

d $x = -3; \frac{-24}{x} + 7 = 5x$

e $x = \frac{1}{4}; x + \frac{2}{3} = \frac{11}{12}$

4 Complete the following equation such that $x = -7$ is the only solution.

$$2(x - 12) + 9 = 3x - \square$$

5 Complete the following equation such that it has no solutions.

$$80x + \frac{1}{6} = \square(8x + 2)$$

6 Solve the following equations:

a $8k + 2 = 15k - 12$

b $8r - 2 = 4r + 14$

c $10r + 8 = 74 - r$

d $12r + 8 = 116 - 6r$

e $36 - 2k = 6k + 20$

f $-90 - 8v = -17v + 9$

g $\frac{14k - 76}{2} - 2 = 2k$

h $-n - \frac{1}{3} = 2$

i $\frac{y - 4.3}{-4} = 7.7$

7 Solve for x :



a $6x + 9 = 10x - 3$

c $\frac{11x + 22}{8} = 33$

e $5x + 2 = 3x + 22$

g $\frac{5x}{6} + 4 = 21.5$

i $\frac{11.8x + 31.86}{4} = -106.2$

k $x - \frac{1}{5} = \frac{3}{4}$

m $8x - \frac{112}{3} = 4x$

o $12x + 19 = x + 85$

b $-\frac{6x}{5} + 5 = 35$

d $3x - 8 = x$

f $-107 - 8x = -17x + 19$

h $6x - \frac{11}{7} = \frac{409}{7}$

j $\frac{4x - 20}{7} = \frac{28}{3}$

l $7x - \frac{90}{7} = x$

n $x - \frac{5}{12} = -\frac{1}{4}$

p $9x + 4 = 10x + 2$

8 Find the value of x .

a $3(x + 1) + 2 = -1$

c $3(x + 6) + 3(x + 24) = 12$

e $-7(x + 1) = -28$

g $3.8(x + 4) - 2 = 1.8$

i $5.3(3x + 55) = 4.9(x + 55)$

k $5(2x + 2) = -3(4x - 5) + 5x$

b $-2(x + 2) - 4 = 4$

d $2(2x + 5) = 3(x + 5)$

f $3(x - 1) - 5 = -8.6$

h $3\left(x + \frac{3}{2}\right) + 3 = -\frac{15}{2}$

j $4(5x + 1) = -3(5x - 5)$

9 Determine the number of solution each of the following equations has.

a $18 + 3x = 3x - 16$

b $8x - 4x = 4$

c $x + 8 = x + 6$

d $2x - 2 = 2x - 2$

e $7x + 4 = 6x + 5$

f $8x = 8x + 8$

g $6x + 5 = 7x - 5$

h $9x - 9 = 9x + 7$

i $7x + 7 = 9x - 7$

j $2x = 2x + 4$



l $10(5 + x) = 10(x + 5)$

m $\frac{10 + x}{10} = \frac{x + 4}{10}$

n $7(7 + x) = 2x + 28$

o $-31 - 3(x - 9) = 2(x - 2) - 5x$

p $24 + 7(x - 23) + 3 = 16 + 8(x + 14) - x - 262$

10 Consider the equation $3x - 29 = 9x + 13$.

- a** Solve for x .
- b** How many solutions does the equation $3x - 29 = 9x + 13$ have?

11 For each of the following relations:

- i** Using n to represent the unknown number, form an equation that represents this relation.
 - ii** Determine the number of solutions the relationship has.
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- a** Three more than three times a number is equal to six more than four times the number
 - b** Six more than two times a number is equal to five more than two times the number
 - c** Two more than four times a number is equal to two added to quadruple the number